

Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 Enzymes are biological catalysts.

(a) Enzyme molecules are formed from chains of amino acids joined together.

Which pair of elements is found in all amino acids?

(1)

- ☐ **A** carbon and oxygen
- ☐ **B** carbon and sulfur
- ☐ **C** nitrogen and magnesium
- ☐ **D** oxygen and magnesium

(b) Which row of the table describes how two amino acids are joined together in an enzyme molecule?

(1)

	Name of bond	Reaction that forms the bond
☐ A	ester	condensation
☐ B	ester	hydrolysis
☐ C	peptide	condensation
☐ D	peptide	hydrolysis



(c) Describe how a chain of amino acids can form an enzyme molecule.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 1 = 5 marks)



P 7 5 7 7 1 A 0 3 2 8

